

The Derivative

Real Analysis

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Definition

Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. A real number L is said to be the **derivative of f at c** if given any $\varepsilon > 0$ there exists $\delta > 0$ such that if $x \in I$ satisfies $0 < |x - c| < \delta$, then

$$\left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon.$$

In this case, we say that f is **real-differentiable at c** and we write $f'(c)$ for L .

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If $f : I \rightarrow \mathbb{R}$ has a derivative at $c \in I$, then f is continuous at c .

Theorem

Let $I \subseteq \mathbb{R}$ be an interval, let $c \in I$, and let $f : I \rightarrow \mathbb{R}$ and $g : I \rightarrow \mathbb{R}$ be functions that are differentiable at c . Then:

a) If $\alpha \in \mathbb{R}$, then the function αf is differentiable at c , and

$$(\alpha f)'(c) = \alpha f'(c).$$

b) The function $f + g$ is differentiable at c , and

$$(f + g)'(c) = f'(c) + g'(c).$$

c) The function fg is differentiable at c , and

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

d) If $g(c) \neq 0$, then the function f/g is differentiable at c , and

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Corollary

If $f_1, f_2, \dots, f_n$ are functions from an interval I to $\mathbb{R}$ that are differentiable at $c \in I$, then:

a) The function $f_1 + f_2 + \dots + f_n$ is differentiable at c , and

$$(f_1 + f_2 + \dots + f_n)'(c) = f_1'(c) + f_2'(c) + \dots + f_n'(c).$$

b) The function $f_1 f_2 \dots f_n$ is differentiable at c , and

$$\begin{aligned}(f_1 f_2 \dots f_n)'(c) &= f_1'(c) f_2(c) \dots f_n(c) + f_1(c) f_2'(c) \dots f_n(c) \\ &\quad + \dots + f_1(c) f_2(c) \dots f_n'(c).\end{aligned}$$

Theorem (Carathéodory's Theorem)

Let f be defined on an interval I that contains the point c . Then f is differentiable at c if and only if there exists a function φ on I that is continuous at c and satisfies

$$f(x) - f(c) = \varphi(x)(x - c) \quad \text{for } x \in I.$$

In this case, we have $\varphi(c) = f'(c)$.

Theorem (Chain Rule)

Let I, J be intervals in $\mathbb{R}$, let $g : I \rightarrow \mathbb{R}$ and $f : J \rightarrow \mathbb{R}$ be functions such that $f(J) \subseteq I$, and let $c \in J$. If f is differentiable at c and if g is differentiable at $f(c)$, then the composite function $g \circ f$ is differentiable at c and

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c).$$

Theorem

Let I be an interval in $\mathbb{R}$ and let $f : I \rightarrow \mathbb{R}$ be strictly monotone and continuous on I . Let $J := f(I)$ and let $g : J \rightarrow \mathbb{R}$ be the strictly monotone and continuous inverse function of f . If f is differentiable at $c \in I$ and if $f'(c) \neq 0$, then g is differentiable at $d := f(c)$ and

$$g'(d) = \frac{1}{f'(c)} = \frac{1}{f'(g(d))}.$$

Theorem

Let I be an interval and let $f : I \rightarrow \mathbb{R}$ be strictly monotone on I . Let $J := f(I)$ and let $g : J \rightarrow \mathbb{R}$ be the inverse function of f . If f is differentiable on I and $f'(x) \neq 0$ for $x \in I$, then g is differentiable on J and

$$g' = \frac{1}{f' \circ g}.$$

Definition

A function $f : I \rightarrow \mathbb{R}$ is said to have a **relative maximum** (or a **relative minimum**) at $c \in I$ if there exists a neighborhood $V := V_\delta(c)$ of c such that $f(x) \leq f(c)$ (or $f(c) \leq f(x)$) for all x in $V \cap I$. We say that f has a **relative extremum** at $c \in I$ if it has a relative maximum or a relative minimum at c .

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Theorem (Interior Extremum Theorem)

Let c be an interior point of the interval I at which $f : I \rightarrow \mathbb{R}$ has a relative extremum. If the derivative of f at c exists, then $f'(c) = 0$.

Corollary

Let $f : I \rightarrow \mathbb{R}$ be continuous on an interval I and suppose that f has a relative extremum at an interior point c of I . Then either the derivative of f at c does not exist or it equals zero.

Theorem (Rolle's Theorem)

Suppose that f is continuous on a closed interval $I := [a, b]$, that the derivative f' exists at every point of the open interval (a, b) , and that $f(a) = f(b) = 0$. Then there exists at least one point c in (a, b) such that $f'(c) = 0$.

Theorem (Mean Value Theorem)

Suppose that f is continuous on a closed interval $I := [a, b]$ and that f has a derivative at every point of the open interval (a, b) . Then there exists at least one point c in (a, b) such that

$$f(b) - f(a) = f'(c)(b - a).$$

Theorem

Suppose that f is continuous on a closed interval $I := [a, b]$, that f is differentiable on the open interval (a, b) , and that $f'(x) = 0$ for $x \in (a, b)$. Then f is constant on I .

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Corollary

Suppose that f and g are continuous on $I := [a, b]$, that they are differentiable on (a, b) , and that $f'(x) = g'(x)$ for all $x \in (a, b)$. Then there exists a constant C such that $f = g + C$ on I .

Theorem

Let $f : I \rightarrow \mathbb{R}$ be differentiable on the interval I . Then:

- a) f is increasing on I if and only if $f'(x) \geq 0$ for all $x \in I$.*
- b) f is decreasing on I if and only if $f'(x) \leq 0$ for all $x \in I$.*

Theorem (First Derivative Test)

Let f be continuous on the interval $I := [a, b]$ and let c be an interior point of I . Suppose that f is differentiable on (a, c) and (c, b) . Then:

- a) If there exists a neighborhood $(c - \delta, c + \delta) \subseteq I$ such that $f'(x) \geq 0$ for $c - \delta < x < c$ and $f'(x) \leq 0$ for $c < x < c + \delta$, then f has a relative maximum at c .
- b) If there exists a neighborhood $(c - \delta, c + \delta) \subseteq I$ such that $f'(x) \leq 0$ for $c - \delta < x < c$ and $f'(x) \geq 0$ for $c < x < c + \delta$, then f has a relative minimum at c .

Lemma

Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, let $c \in I$, and suppose that f has a derivative at c . Then:

- a) If $f'(c) > 0$, then there exists a number $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ such that $c < x < c + \delta$.
- b) If $f'(c) < 0$, then there exists a number $\delta > 0$ such that $f(x) < f(c)$ for $x \in I$ such that $c - \delta < x < c$.

Theorem (Darboux's Theorem)

If f is differentiable on $I := [a, b]$ and if k is a number between $f'(a)$ and $f'(b)$, then there exists at least one point c in (a, b) such that $f'(c) = k$.

Theorem

Let f, g be defined on $[a, b]$, let $f(a) = g(a) = 0$, and let $g(x) \neq 0$ for $a < x < b$. If f and g are differentiable at a and if $g'(a) \neq 0$, then the limit of f/g at a exists and equals $f'(a)/g'(a)$. Therefore

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

Theorem (Cauchy's Mean Value Theorem)

Let f, g be continuous on $[a, b]$ and differentiable on (a, b) , and suppose that $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Theorem (L'Hôpital's Rule I)

Let $-\infty \leq a < b \leq \infty$ and let f, g be differentiable on (a, b) such that $g'(x) \neq 0$ for all $x \in (a, b)$. Suppose that

$$\lim_{x \rightarrow a^+} f(x) = 0 = \lim_{x \rightarrow a^+} g(x).$$

a) If $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}$, then $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$.

b) If $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \in \{-\infty, \infty\}$, then $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$.

Theorem (L'Hôpital's Rule II)

Let $-\infty \leq a < b \leq \infty$ and let f, g be differentiable on (a, b) such that $g'(x) \neq 0$ for all $x \in (a, b)$. Suppose that

$$\lim_{x \rightarrow a^+} g(x) = \pm\infty.$$

a) If $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}$, then $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$.

b) If $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \in \{-\infty, \infty\}$, then $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$.

Theorem (Taylor's Theorem)

Let $n \in \mathbb{N}$, let $I := [a, b]$, and let $f : I \rightarrow \mathbb{R}$ be such that f and its derivatives $f', f'', \dots, f^{(n)}$ are continuous on I and such that $f^{(n+1)}$ exists on (a, b) . If $x_0 \in I$, then for any $x \in I$ there exists a point c between x and x_0 such that

$$\begin{aligned} f(x) = & f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 \\ & + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}. \end{aligned}$$

Theorem

Let I be an interval, let x_0 be an interior point of I , and let $n \geq 2$. Suppose that the derivatives $f', f'', \dots, f^{(n)}$ exist and are continuous in a neighborhood of x_0 and that $f'(x_0) = \dots = f^{(n-1)}(x_0) = 0$, but $f^{(n)}(x_0) \neq 0$.

- i) If n is even and $f^{(n)}(x_0) > 0$, then f has a relative minimum at x_0 .
- ii) If n is even and $f^{(n)}(x_0) < 0$, then f has a relative maximum at x_0 .
- iii) If n is odd, then f has neither a relative maximum nor a relative minimum at x_0 .

Definition

Let $I \subseteq \mathbb{R}$ be an interval. A function $f : I \rightarrow \mathbb{R}$ is said to be **convex on I** if for any t satisfying $0 \leq t \leq 1$ and any points $x_1, x_2 \in I$ we have

$$f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2).$$

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Let I be an open interval and let $f : I \rightarrow \mathbb{R}$ have a second derivative on I . Then f is a convex function on I if and only if $f''(x) \geq 0$ for all $x \in I$.